Data Architecture - Network Request Automation

1. Description

The Automated Network Request Management solution in ServiceNow is designed using a structured and scalable data architecture that integrates standard ServiceNow platform tables with a custom-built Network Database table. This architecture ensures efficient data storage, secure access control, seamless workflow automation, and accurate reporting. Network requests are captured through Service Catalog submissions and stored using well-defined fields such as request type, access level, device details, business justification, requester information, and approval status. Dynamic form behavior using UI policies and UI logic ensures that only relevant fields are displayed, improving user experience and data quality. List views and form layouts provide real-time operational visibility for approvers and network teams, enabling effective monitoring, tracking, and decision-making throughout the request lifecycle.

2. Data Model Overview

The data architecture is based on the following key components:

- Service Catalog Tables – Capture user request inputs
- Custom Network Database Table – Stores structured network request data
- Approval Tables – Manage approval workflow
- Task Tables – Track execution activities
- System Tables – Support notifications, auditing, and reporting

This architecture ensures data consistency, security, scalability, and automation readiness.

3. Table Design – Network Database

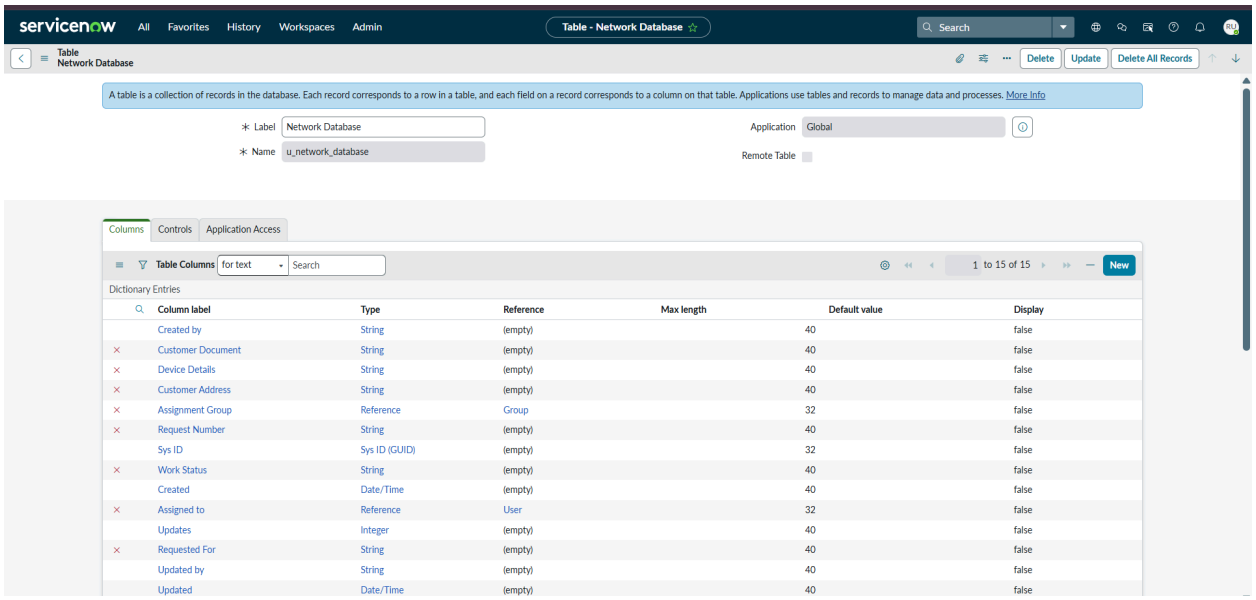
A custom table named Network Database is created to act as the central repository for all network-related requests.

This table stores:

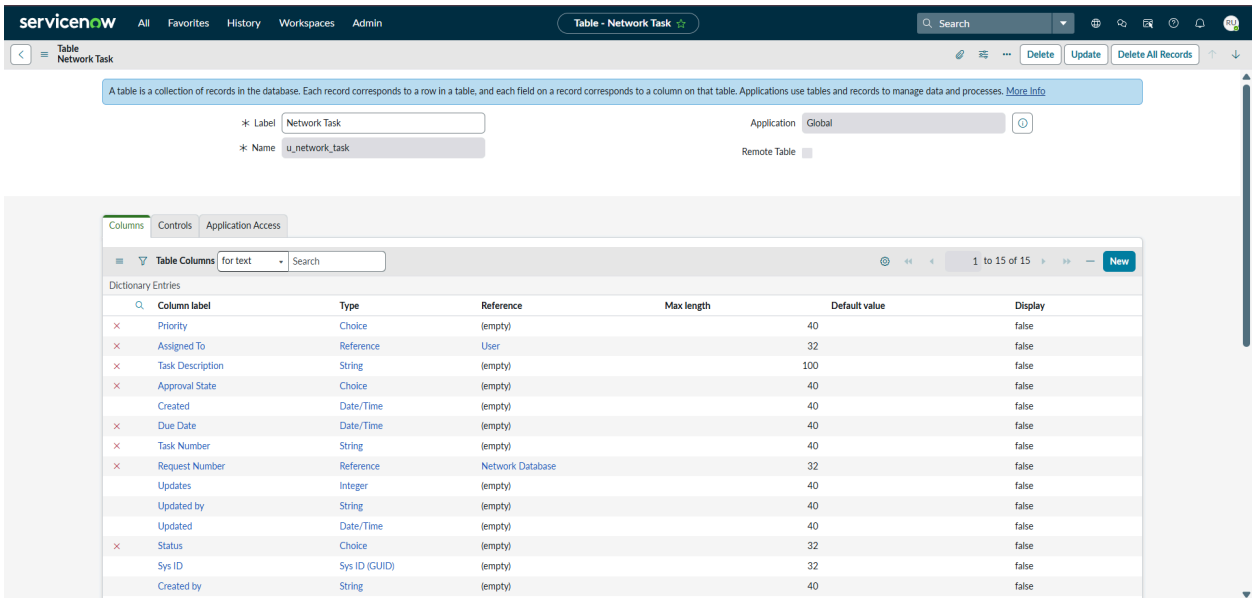
- Request Type
- Access Level
- Device Information
- Business Justification
- Requester Details
- Approval Status

- Task Status
- Timestamps

This structured table design enables accurate tracking, auditing, and reporting of all network operations.



Column label	Type	Reference	Max length	Default value	Display
Created by	String	(empty)	40		false
Customer Document	String	(empty)	40		false
Device Details	String	(empty)	40		false
Customer Address	String	(empty)	40		false
Assignment Group	Reference	Group	32		false
Request Number	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Work Status	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Assigned to	Reference	User	32		false
Updates	Integer	(empty)	40		false
Requested For	String	(empty)	40		false
Updated by	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false



Column label	Type	Reference	Max length	Default value	Display
Priority	Choice	(empty)	40		false
Assigned To	Reference	User	32		false
Task Description	String	(empty)	100		false
Approval State	Choice	(empty)	40		false
Created	Date/Time	(empty)	40		false
Due Date	Date/Time	(empty)	40		false
Task Number	String	(empty)	40		false
Request Number	Reference	Network Database	32		false
Updates	Integer	(empty)	40		false
Updated by	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Status	Choice	(empty)	32		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false

4. Field Architecture & Data Modeling

Each field in the Network Database table is carefully designed using appropriate data types to ensure data accuracy, integrity, and performance.

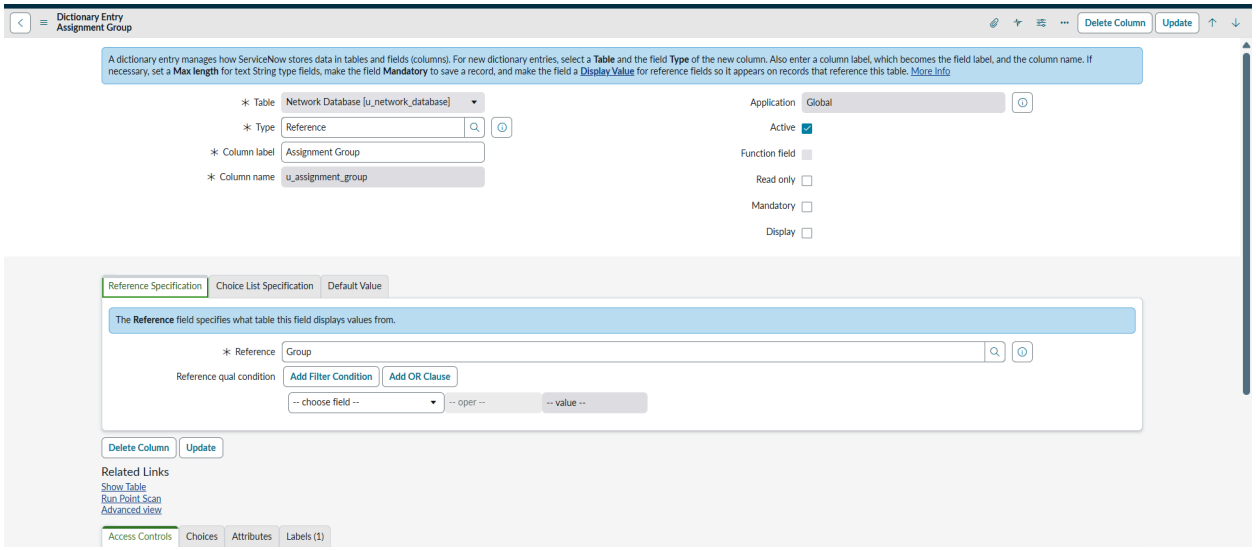
Common Field Types Used

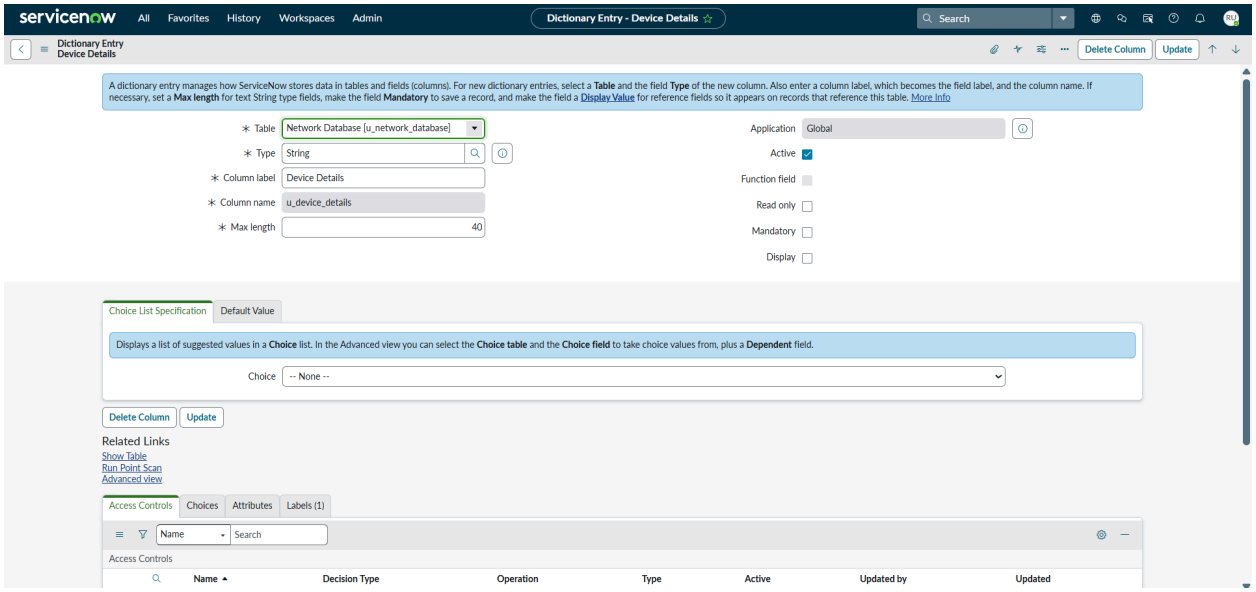
- **String:** For short textual data (e.g., device name, business justification)
- **Choice:** For controlled dropdown selections (e.g., request type, access level, status)
- **Reference:** For linking records with system tables (e.g., user, department, approver)
- **Boolean:** For true/false indicators
- **Date / Date-Time:** For request tracking and audit timestamps

Key Design Principles

- Mandatory fields enforce data completeness
- Reference fields ensure relational integrity
- Choice fields maintain data standardization
- Read-only fields protect system-generated values

This ensures high-quality data capture, reduced errors, and consistent reporting.





5. Data Flow Architecture

1. User submits the Service Catalog form.
2. Data is captured using catalog variables.
3. Records are created in the Network Database table.
4. Approval workflow updates approval state fields.
5. Task creation updates execution status fields.
6. Final status updates allow reporting and auditing.

6. Benefits of the Data Architecture

- Centralized data storage
- Structured and scalable design
- High data accuracy and consistency
- Improved operational visibility
- Secure access control
- Seamless integration with workflow automation
- Reliable reporting and auditing

7. Conclusion

The Data Architecture of the Network Request Automation solution provides a robust, scalable, and secure foundation for managing network service requests in ServiceNow. By combining structured data modeling, dynamic UI behavior, and automated workflows, the system ensures efficient operations, improved user experience, and high-quality data management.